

Local Ollama + VPS Setup

Run the AI model on your Windows PC GPU and host the website/backend on your VPS.

Installed model: qwen3.5:4b. It runs in Docker on your PC.

1. Windows PC: install Docker Desktop

Open PowerShell as Administrator:

```
ws1 --update
winget install -e --id Docker.DockerDesktop
```

Restart if Docker requests it. Open Docker Desktop and wait for *Engine running*.

```
docker version
```

2. Windows PC: start Ollama with GPU

```
docker volume create ollama

docker run -d `
  --name ollama `
  --restart unless-stopped `
  --gpus all `
  -e OLLAMA_KEEP_ALIVE=30m `
  -e OLLAMA_NUM_PARALLEL=1 `
  -e OLLAMA_MAX_LOADED_MODELS=1 `
  -v ollama:/root/.ollama `
  -p 127.0.0.1:11434:11434 `
  ollama/ollama

docker ps
```

3. Windows PC: install and test the model

```
docker exec -it ollama ollama pull qwen3.5:4b
docker exec -it ollama ollama list
docker exec -it ollama ollama run qwen3.5:4b "Say hello in one short sentence."
docker exec ollama ollama ps
nvidia-smi
```

Do not use: qwen3.5:7b; that tag does not exist. For your 8 GB GPU and fast replies, use qwen3.5:4b.

4. Private PC-to-VPS connection with Tailscale

Install Tailscale on the Windows PC, sign in, then run PowerShell as Administrator:

```
tailscale status
tailscale serve --bg --http=11434 http://127.0.0.1:11434
tailscale serve status
```

Your current private model address:

```
http://desktop-2ram9tm.tail48accd.ts.net:11434
```

Never use Tailscale Funnel and never open router port 11434. Tailscale Serve keeps Ollama private to your tailnet.

5. VPS: install Tailscale and test Ollama

```
curl -fsSL https://tailscale.com/install.sh | sh
sudo tailscale up

curl http://desktop-2ram9tm.tail48accd.ts.net:11434/api/tags
```

The last command must list qwen3.5:4b.

6. VPS: connect the running gateway

```
cd /var/www/ollama-gateway
cp docker-compose.yml docker-compose.yml.before-local-ollama

sed -i 's|OLLAMA_BASE_URL: http://ollama:11434|OLLAMA_BASE_URL: http://desktop-2ram9tm.tail48accd.ts.net:11434|' docker-compose.yml

grep -n 'OLLAMA_BASE_URL' docker-compose.yml
docker compose up -d --no-deps --force-recreate gateway
curl http://127.0.0.1:8000/health/ready
```

Health must return "ollama":true.

7. Fast replies

1. Keep OLLAMA_NUM_PARALLEL=1 and OLLAMA_MAX_LOADED_MODELS=1 on the PC.
2. Use streaming in the website so users see tokens immediately.
3. Disable thinking for ordinary chat requests by sending "think": false to Ollama.
4. Use a context window around 4096 and cap response length for normal chat.

```
time curl -s http://desktop-2ram9tm.tail48accd.ts.net:11434/api/generate \
-d '{"model":"qwen3.5:4b","prompt":"Reply only: OK","stream":false,"think":false,"options":
{"num_predict":16,"num_ctx":4096}}'
```

First request can be slower while the model loads into GPU memory. Do not press Ctrl+C while it is loading.